# Global Patent Intelligence Data Pipeline

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In this project, you will build a simple data system that collects, cleans, stores, and analyzes **real-world patent data**.

Patents are official records of inventions. Companies and governments use them to understand:

- New technologies
- Top inventors
- Leading companies
- Innovation trends across countries

You will act like a **data engineer** and build a small version of this system.

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## ◇ Data source: PatentsView Granted Patent Disambiguated Data

Link: https://data.uspto.gov/bulkdata/datasets/pvgpatdis?fileDataFromDate=1976-01-01&fileDataToDate=2025-09-30

You will use this to get patent data such as:

(Be sure read the accompanying **PV_grant_data_dictionary.pdf** on the same page)

- Patent titles
- Abstracts
- Inventors
- Companies (assignees)
- Dates
- Classification

You will build a simple data pipeline:

Data Source (Files)
↓
Python Script (Reading the Data)
↓
Clean Data using pandas
↓
Store in SQL Database
↓
Analyze using SQL Queries
↓
Create Reports (CSV, JSON, Console)

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You will use:

## Python

- Getting data from the source
- Writing scripts

## pandas

- Cleaning messy data
- Fixing missing values
- Organizing tables

## SQL

- Storing data in tables
- Writing queries
- Joining tables

## GitHub

- Saving your work
- Tracking changes

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Your database must include these tables (and any others you find fit):

## 📌 patents

- patent_id
- title
- abstract
- filing_date
- year

## 📌 inventors

- inventor_id
- name
- country

## 📌 companies (assignees)

- company_id
- name

## 📌 relationships table

- patent_id
- inventor_id
- company_id

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Your SQL queries must return:

## Q1: Top Inventors

Who has the most patents?

## Q2: Top Companies

Which companies own the most patents?

## Q3: Countries

Which countries produce the most patents?

## Q4: Trends Over Time

How many patents are created each year?

## Q5: JOIN Query

Combine patents with inventors and companies

## Q6: CTE Query (WITH statement)

Break a complex query into steps

## Q7: Ranking Query

Rank inventors using window functions

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You will submit a GitHub project with:

## 🗀 Code Files

- Python scripts
- SQL scripts

## 🗀 Clean Data Files

- clean_patents.csv
- clean_inventors.csv
- clean_companies.csv

## 🗀 Database File

- schema.sql

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You MUST create 3 types of reports:

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## 🥇 A. Console Report (Terminal Output)

Example:

========== PATENT REPORT ==========

Total Patents: 100,000

Top Inventors: 1. John Smith - 312 2. Alice Johnson - 290

Top Companies: 1. IBM - 12,000

Top Countries: 1. USA 2. China

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You must export at least 2 files:

- top_inventors.csv
- top_companies.csv
- country_trends.csv

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JSON Report. Example:

{ “total_patents”: 100000, “top_inventors”: [ {“name”: “John Smith”, “patents”: 312} ], “top_companies”: [ {“name”: “IBM”, “patents”: 12000} ], “top_countries”: [ {“country”: “USA”, “share”: 0.65} ] }

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You can get extra marks for:

- Data visualizations (graphs)
- Simple dashboard (Streamlit)
- Advanced analysis of patent categories

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Your project must be reproducible:

👉 Anyone should be able to clone your GitHub repo and run your code to get the same results.

By the end of this project, you will be able to:

- Work with real API data
- Clean and process datasets
- Store data in SQL databases
- Write analytical SQL queries
- Generate structured reports
- Build a full data pipeline like a real data engineer